Chapter 5: Differentiation

Section: Differentiation of Vector-Valued Functions

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Motivation

Beyond Real-Valued Functions

- So far in Chapter 5: $f : [a, b] \to \mathbb{R}$ — derivative, mean value theorems, L'Hospital's rule, Taylor's theorem

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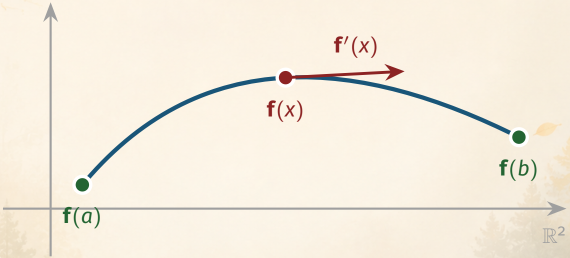
- Many natural functions are vector-valued: complex $f = f_1 + if_2$, curves $\mathbf{f} : [a, b] \to \mathbb{R}^k$
- Central question: which results survive?
    - definitions and algebra: *yes*
    - mean value theorem, L'Hospital's rule: no!
    - the rescue: *Theorem 5.19*, a mean value inequality

The Derivative as a Velocity Vector



A curve $\mathbf{f}\colon [a, b] \to \mathbb{R}^2$; its derivative $\mathbf{f}'(x)$ is the velocity vector of the motion.

Vector-Valued Derivatives

Remarks 5.16: The Complex Case

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Definition 5.1 applies without any change to complex functions f on $[a, b]$, and Theorems 5.2 and 5.3, together with their proofs, remain valid.

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Real and imaginary parts

If $f(t) = f_1(t) + if_2(t)$ with f_1, f_2 real, then

$$f'(x) = f_1'(x) + if_2'(x); \tag{29}$$

f is differentiable at $x \iff$ both f_1 and f_2 are differentiable at x.

The Vector Case: f Maps $[a, b]$ into $\mathbb{R}^k$

The derivative of a vector-valued function

Apply Definition 5.1 verbatim: the difference quotient

$$\phi(t) = \frac{\mathbf{f}(t) - \mathbf{f}(x)}{t - x}$$

is now, for each t, a point in $\mathbb{R}^k$. Then $\mathbf{f}'(x)$ is that point of $\mathbb{R}^k$ (if there is one) for which

$$\lim_{t \to x} \left| \frac{\mathbf{f}(t) - \mathbf{f}(x)}{t - x} - \mathbf{f}'(x) \right| = 0. \tag{30}$$

$\mathbf{f}'$ is again a function with values in $\mathbb{R}^k$.

Componentwise Differentiation

Components (as in Theorem 4.10)

If $f_1, \ldots, f_k$ are the components of $\mathbf{f}$, then

$$\mathbf{f}' = (f_1', \ldots, f_k'), \tag{31}$$

and $\mathbf{f}$ is differentiable at $x \iff$ each of $f_1, \ldots, f_k$ is differentiable at x.

Consequence: for *definitions and algebra*, vector calculus reduces to k real problems.

What Survives

- Theorem 5.2: differentiable at $x \Rightarrow$ continuous at x
- Theorem 5.3(a): $(\mathbf{f} + \mathbf{g})'(x) = \mathbf{f}'(x) + \mathbf{g}'(x)$

What Survives

- Theorem 5.2: differentiable at $x \Rightarrow$ continuous at x
- Theorem 5.3(a): $(\mathbf{f} + \mathbf{g})'(x) = \mathbf{f}'(x) + \mathbf{g}'(x)$
- Theorem 5.3(b), with fg replaced by the inner product (Def. 4.3):

$$(\mathbf{f} \cdot \mathbf{g})'(x) = \mathbf{f}'(x) \cdot \mathbf{g}(x) + \mathbf{f}(x) \cdot \mathbf{g}'(x)$$

Where the Story Turns

- The mean value theorem (5.10) *fails* for complex-valued functions (Example 5.17)
- L'Hospital's rule (5.13) *fails* for complex-valued functions (Example 5.18)

Already for $k = 2$, equality-type theorems break down.

The Mean Value Theorem Fails

Example 5.17: A Trip Around the Circle

Example 5.17

Define, for real x,

$$f(x) = e^{ix} = \cos x + i \sin x. \tag{32}$$

- Take (32) as the *definition* of e^{ix} (full discussion in Chap. 8)
- As x increases, $f(x)$ moves along the unit circle

Example 5.17: Zero Displacement, Unit Speed

The displacement over $[0, 2\pi]$

$$f(2\pi) - f(0) = 1 - 1 = 0 \tag{33}$$

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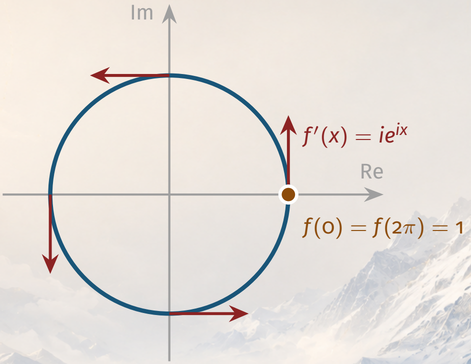
Theorem 5.10 would require $f(2\pi) - f(0) = 2\pi f'(x)$, i.e. $f'(x) = 0$ for some x — impossible. Thus Theorem 5.10 fails for complex functions.

The Picture: Motion on the Unit Circle



Displacement over $[0, 2\pi]$: zero.   Speed at every instant: one.

L’Hospital’s Rule Fails

Example 5.18: The Setup

Example 5.18

On the segment $(0,1)$, define $f(x) = x$ and

$$g(x) = x + x^2 e^{i/x^2}. \tag{35}$$

Since $|e^{it}| = 1$ for all real t: $\dfrac{f(x)}{g(x)} = \dfrac{1}{1 + x\, e^{i/x^2}}$ and $\left|x\, e^{i/x^2}\right| = x \to 0$, so

$$\lim_{x \to 0} \frac{f(x)}{g(x)} = 1. \tag{36}$$

Example 5.18: The Derivative of g Blows Up

Differentiate (35)

$$g'(x) = 1 + \left\{ 2x - \frac{2i}{x} \right\} e^{i/x^2} \qquad (0 < x < 1), \tag{37}$$

so that

$$|g'(x)| \geq \left| 2x - \frac{2i}{x} \right| - 1 \geq \frac{2}{x} - 1. \tag{38}$$

In particular $g'(x) \neq 0$ on $(0, 1)$, and $|g'(x)| \to \infty$ as $x \to 0$.

Example 5.18: The Punch Line

The ratio of derivatives

Since $f'(x) = 1$, (38) gives

$$\left|\frac{f'(x)}{g'(x)}\right| = \frac{1}{|g'(x)|} \leq \frac{x}{2-x}, \tag{39}$$

and so

$$\lim_{x\to 0} \frac{f'(x)}{g'(x)} = 0. \tag{40}$$

By (36) and (40): $\dfrac{f}{g} \to 1$ but $\dfrac{f'}{g'} \to 0$. L'Hospital's rule fails.

What Went Wrong?

The mechanism behind both failures

- The perturbation x^2e^{i/x^2} is small: $\left|x^2e^{i/x^2}\right| = x^2 \to 0$
- Its derivative is huge: size $\approx 2/x \to \infty$: ever-faster *spinning*
- A vector-valued function can move at high speed while *staying close to a point*

The Inequality That Survives

Salvaging the Mean Value Theorem

A consequence of Theorem 5.10 (real case)

$$|f(b) - f(a)| \leq (b - a) \sup_{a<x<b} |f'(x)| \tag{41}$$

- For applications, (41) is *almost as useful* as Theorem 5.10 itself
- And — unlike (5.10) — it remains true for vector-valued functions

Theorem 5.19: The Mean Value Inequality

Theorem 5.19

Suppose $\mathbf{f}$ is a continuous mapping of $[a, b]$ into $\mathbb{R}^k$ and $\mathbf{f}$ is differentiable in (a, b). Then there exists $x \in (a, b)$ such that

$$|\mathbf{f}(b) - \mathbf{f}(a)| \leq (b - a)\,|\mathbf{f}'(x)|.$$

Check against Example 5.17: $|f(2\pi) - f(0)| = 0 \leq 2\pi \cdot 1$ (inequality, not equality!)

Proof Strategy

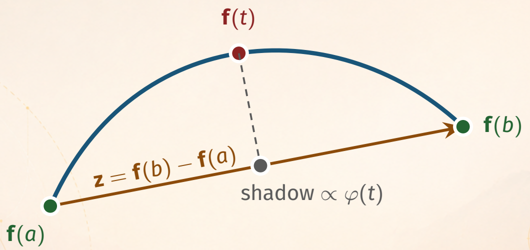
Goal: find $x \in (a, b)$ with $|\mathbf{f}(b) - \mathbf{f}(a)| \le (b - a)\,|\mathbf{f}'(x)|$.

Approach:

1. Let $\mathbf{z} = \mathbf{f}(b) - \mathbf{f}(a)$ be the displacement vector
2. Project the motion onto $\mathbf{z}$: $\varphi(t) = \mathbf{z} \cdot \mathbf{f}(t)$ is real-valued
3. Apply the real mean value theorem to φ
4. Finish with the Schwarz inequality

Technique: reduction to the real case by projection.

The Projection Idea



$\varphi(t) = \mathbf{z} \cdot \mathbf{f}(t)$: a real function tracking progress along the chord.

Proof of Theorem 5.19 — Step 1: Project onto the Chord

Step 1: A real-valued auxiliary function

Put $\mathbf{z} = \mathbf{f}(b) - \mathbf{f}(a)$, and define

$$\varphi(t) = \mathbf{z} \cdot \mathbf{f}(t) \qquad (a \le t \le b).$$

Then φ is real-valued, continuous on $[a, b]$, differentiable in (a, b), with $\varphi'(t) = \mathbf{z} \cdot \mathbf{f}'(t)$.

Proof of Theorem 5.19 — Step 2: Apply the Real MVT

Step 2: The mean value theorem for φ

By Theorem 5.10, there is a point $x \in (a, b)$ with

$$\varphi(b) - \varphi(a) = (b - a)\,\varphi'(x) = (b - a)\,\mathbf{z} \cdot \mathbf{f}'(x).$$

Proof of Theorem 5.19 — Step 3: Compute the Increment

Step 3: The increment of φ is $|\mathbf{z}|^2$

On the other hand,

$$\varphi(b) - \varphi(a) = \mathbf{z} \cdot \mathbf{f}(b) - \mathbf{z} \cdot \mathbf{f}(a) = \mathbf{z} \cdot \big(\mathbf{f}(b) - \mathbf{f}(a)\big) = \mathbf{z} \cdot \mathbf{z} = |\mathbf{z}|^2.$$

Proof of Theorem 5.19 — Step 4: Schwarz Inequality

Step 4: Conclusion

The Schwarz inequality now gives

$$|\mathbf{z}|^2 = (b-a)\,|\mathbf{z}\cdot\mathbf{f}'(x)| \leq (b-a)\,|\mathbf{z}|\,|\mathbf{f}'(x)|.$$

If $\mathbf{z} \neq \mathbf{0}$, divide by $|\mathbf{z}|$ (if $\mathbf{z} = \mathbf{0}$, the conclusion is trivial):

$$\boxed{|\mathbf{f}(b) - \mathbf{f}(a)| \leq (b-a)\,|\mathbf{f}'(x)|}$$

This proves Theorem 5.19. ■

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Key takeaways from this section:

- **The derivative extends verbatim** to complex and $\mathbb{R}^k$-valued functions, and works componentwise: $\mathbf{f}' = (f_1', \dots, f_k')$; Theorems 5.2 and 5.3(a),(b) (inner product) survive

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- **Two famous results fail:** the MVT ($f(x) = e^{ix}$: zero displacement, unit speed) and L'Hospital's rule ($g(x) = x + x^2 e^{i/x^2}$: $f/g \to 1$ but $f'/g' \to 0$)

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- **Two famous results fail:** the MVT ($f(x) = e^{ix}$: zero displacement, unit speed) and L'Hospital's rule ($g(x) = x + x^2 e^{i/x^2}$: $f/g \to 1$ but $f'/g' \to 0$)
- **Theorem 5.19 saves the day:** $|\mathbf{f}(b) - \mathbf{f}(a)| \leq (b-a)\,|\mathbf{f}'(x)|$ — proved by projection onto the chord + MVT + Schwarz

Coming up next: Chapter 6 — the Riemann–Stieltjes integral $\int_a^b f\,d\alpha$.